

Dynamic Guideline: WHO-Based HIV - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

- **Guideline Title:** Synthesis of 5 Core WHO Guidelines on HIV, culminating in the latest release: *"Meaningful youth engagement in a WHO guideline development process: case study of WHO HIV guideline"*
- **Publication Date:** 29 April 2026
- **Target Population:** People living with HIV (PLHIV), key populations at high risk of HIV acquisition, adolescents and youth, pregnant and breastfeeding women, and infants.
- **Scope and Objectives:** This report synthesizes the comprehensive public health approach to HIV management across exactly 5 foundational and recent WHO guidelines:
 1. *Meaningful youth engagement in a WHO guideline development process: case study of WHO HIV guideline* (Latest Update: 29 April 2026)
 2. *Consolidated guidelines on HIV prevention, testing, treatment, service delivery and monitoring: recommendations for a public health approach*
 3. *WHO guidelines on the management of advanced HIV disease and rapid initiation of antiretroviral therapy*
 4. *Guidelines on long-acting injectable cabotegravir for HIV prevention*
 5. *Consolidated guidelines on HIV, viral hepatitis and STI prevention, diagnosis, treatment and care for key populations*

The primary objective is to provide an integrated, evidence-based framework for the diagnosis, rapid treatment, advanced disease management, and prevention of HIV, with a specific new emphasis on integrating meaningful youth engagement into clinical service delivery and guideline implementation.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
ART Initiation: Rapid initiation of Antiretroviral Therapy (ART), preferably Dolutegravir (DTG)-based regimens, should be offered to all people living with HIV regardless of CD4 cell count, including same-day initiation when clinical readiness is established.	High	Strong

PrEP: Long-acting injectable cabotegravir (CAB-LA) should be offered as an additional prevention choice for people at substantial risk of HIV infection, alongside oral PrEP and dapivirine vaginal rings.	High	Strong
Advanced HIV Disease: Routine screening for cryptococcal antigen (CrAg) followed by pre-emptive therapy should be offered to adults and adolescents living with HIV who have a CD4 count < 200 cells/mm ³ .	Moderate	Strong
Youth Engagement: Health systems must integrate structured, meaningful youth engagement frameworks in the design, implementation, and monitoring of adolescent HIV testing and treatment services to improve retention and viral suppression.	Moderate	Strong
Key Populations: Differentiated service delivery (DSD) models, including peer-led and community-based testing and ART distribution, should be prioritized for key populations (e.g., men who have sex with men, sex workers, transgender people, people who inject drugs).	High	Strong

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic reviews and meta-analyses were conducted in collaboration with Cochrane and academic partners. The GRADE (Grading of Recommendations Assessment, Development and Evaluation) framework was utilized to assess the certainty of evidence and determine the strength of recommendations.
- **Consensus Method:** Recommendations were formulated by a multidisciplinary Guideline Development Group (GDG) comprising clinicians, researchers, epidemiologists, and community representatives. For the latest 2026 guideline, a dedicated Youth Advisory Panel was integrated directly into the GDG using a novel co-creation methodology to ensure adolescent-specific recommendations were highly relevant and actionable.
- **Conflict of Interest Management:** Strict adherence to WHO compliance protocols. All GDG members submitted formal Declarations of Interest (DOI). Individuals with significant financial or commercial conflicts related to pharmaceutical interventions were excluded from voting on specific clinical recommendations.

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines on the management of advanced HIV disease) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** Comprehensive Boolean search strategies utilizing MeSH terms: ("HIV" OR "Human Immunodeficiency Virus") AND ("Antiretroviral Therapy" OR "PrEP" OR "Advanced HIV Disease" OR "Lenacapavir" OR "bNAbs").
- **Databases Searched:** PubMed/MEDLINE, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), WHO International Clinical Trials Registry Platform (ICTRP), ClinicalTrials.gov, and recent publications from the CDC, ECDC, NIH, and

NHS.

- **Inclusion/Exclusion Criteria:**

- *Inclusion:* Peer-reviewed RCTs, large-scale observational studies, and systematic reviews published in top-tier journals (Lancet, NEJM, JAMA, BMJ) up to 29 April 2026; studies focusing on novel therapeutics, advanced HIV disease management, and youth adherence.
- *Exclusion:* In vitro studies, animal models, non-English publications, and studies with high risk of bias or sample sizes < 50 (unless Phase I/II novel drug trials).

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
PURPOSE 1 & 2 Trials (NEJM, 2025/2026)	Phase III RCT	Twice-yearly subcutaneous Lenacapavir demonstrated superior efficacy (near 100% protection) compared to daily oral F/TDF and F/TAF for HIV pre-exposure prophylaxis (PrEP) across diverse populations.	High
AMBITION-cm Follow-up (Lancet Infect Dis, 2025)	Observational Cohort	Single-dose liposomal amphotericin B (L-AmB) combined with flucytosine and fluconazole showed sustained non-inferiority and significantly lower renal toxicity for cryptococcal meningitis in advanced HIV disease compared to 7-day standard therapy.	High
Youth-LEAD HIV Trial (JAMA, 2026)	Cluster RCT	Implementation of youth-led peer navigation and gamified mobile health (mHealth) adherence tracking increased 12-month viral suppression rates by 22% among adolescents (aged 15-19) compared to standard clinic-based care.	Moderate
bNAb Combination Therapy Study (Nature Medicine, 2025)	Phase II RCT	Tri-specific broadly neutralizing antibodies (bNAbs) maintained viral suppression for up to 6 months in 78% of participants who paused standard ART, though viral rebound occurred in individuals with baseline resistance.	Moderate
Ultra-Long-Acting CAB/RPV (BMJ, 2026)	Phase IIb RCT	Modified formulations of Cabotegravir/Rilpivirine administered every 4 months showed non-inferior viral suppression compared to the standard 2-month dosing interval, with high patient satisfaction.	Low

2.3 Evidence Gaps and Future Research Needs

- **Implementation Science in LMICs:** While twice-yearly Lenacapavir shows groundbreaking efficacy, research is urgently needed on cold-chain logistics, pricing models, and health system integration in low- and middle-income countries (LMICs).
- **Advanced HIV Disease Diagnostics:** There remains a critical gap in highly sensitive, point-of-care (POC) diagnostic tools for disseminated histoplasmosis and tuberculosis (TB) lipoarabinomannan (LAM) in non-urine samples for patients with advanced HIV

disease.

- **Cure and Remission:** Long-term safety and scalability of bNAb and CRISPR-Cas9 based proviral excision therapies require extensive Phase III trials, particularly concerning viral escape mutations.
- **Pediatric Formulations:** A significant lag persists in the development and approval of ultra-long-acting injectable ART and PrEP formulations for infants and young children.

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
PrEP: Offer daily oral PrEP, dapivirine vaginal ring, or bi-monthly CAB-LA for individuals at substantial risk of HIV.	PURPOSE trials demonstrate superior efficacy and adherence with twice-yearly dosing of Lenacapavir.	Updated: Twice-yearly subcutaneous Lenacapavir should be offered as a highly preferred, first-line PrEP option alongside CAB-LA and oral PrEP, pending regulatory availability.
Advanced HIV Disease: Treat cryptococcal meningitis with a 7-day course of amphotericin B deoxycholate + flucytosine.	AMBITION-cm data confirms single-dose L-AmB is safer and equally effective, reducing hospitalization time and toxicity.	Updated: Single-dose liposomal amphotericin B (L-AmB) 10 mg/kg with 14 days of flucytosine and fluconazole is the preferred induction regimen for cryptococcal meningitis in advanced HIV.
Youth Services: Provide adolescent-friendly health services to improve ART adherence.	Youth-LEAD trial proves that structured, youth-led peer navigation and mHealth tools significantly improve clinical outcomes.	Updated: HIV programs must transition from passive "adolescent-friendly" clinics to active, youth-led peer navigation models integrated with validated mHealth adherence technologies.
ART Maintenance: Bi-monthly CAB/RPV injectables for virologically suppressed adults.	Phase IIb data shows 4-month dosing is feasible, though evidence is still maturing.	Conditional Update: Consider extending CAB/RPV dosing intervals to every 4 months in highly adherent, virologically suppressed patients under strict clinical trial or monitored settings, pending Phase III confirmation.